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Modul 3: Electrical components

Stand: 18. Februar 2026

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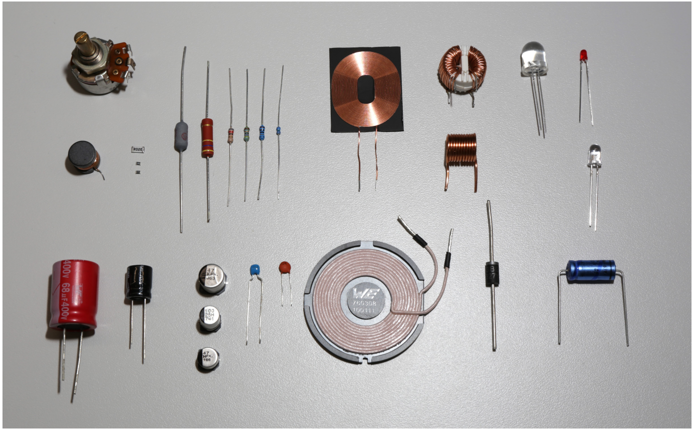

400V 68µF400V

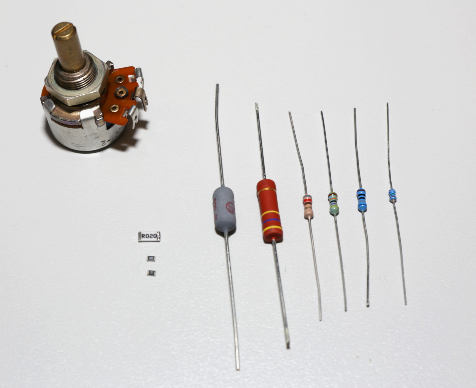
Learning objectives: Conductivity and resistance

Learning objectives: Leitfähigkeit und Widerstand

The Students

- are familiar with the electrical component known as a resistor.
- can explain the different component designs of resistors.
- can perform calculations based on the specific resistance ρ or the specific conductance κ.
- can analyze the differences in conductivity of various materials based on their atomic structure and temperature.

- Different component designs of resistors



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$$J = \frac{\Delta Q \cdot \Delta x}{\Delta t \cdot \Delta V} \tag{4}$$

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Since velocity always has a magnitude and a direction, current density is also a vector.

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In summary, this description results in the description of Ohm's law:

$$\vec{J} = \kappa \cdot \vec{E} \leftrightarrow \vec{E} = \frac{\vec{J}}{\kappa} \tag{13}$$

Electrical conductivity

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$$U_{12} = \int\limits_{0}^{l} \frac{J}{\kappa}\mathrm{d}x \tag{16}$$

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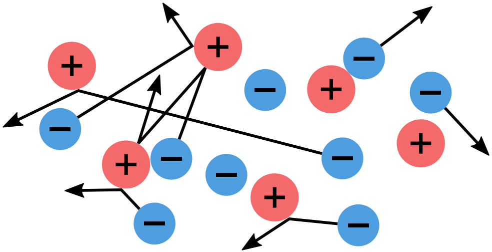
Since $\frac{l}{A \cdot \kappa}$ is the electrical resistance, the following simplification results:

$$U = R \cdot I \tag{22}$$

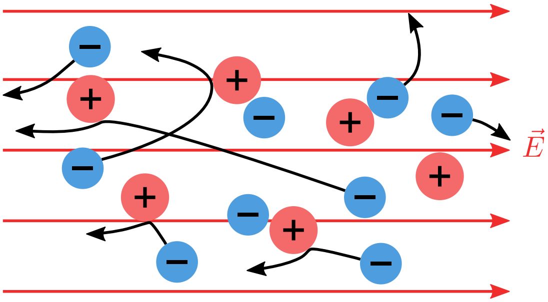
Drift behavior of electrons in conductors

- Unordered motion of free electrons (blue) between positively charged atomic nuclei (red)



Drift behavior of electrons in conductors

- Directed movement of electrons against the direction of the electric field $\vec{E}$



Mathematical relationships of conductivity

$$[\kappa] = 1\ \frac{\mathsf{Siemens}}{\mathsf{meter}} = 1\ \frac{\mathsf{S}}{\mathsf{m}} = 1\ \frac{1}{\Omega \cdot \mathsf{m}}$$

$$\kappa = \frac{1}{\rho_\mathrm{R}}$$

$$\rho_\mathrm{R} = \rho_{20^\circ\mathrm{C}} \cdot (1 + \alpha(\vartheta - 20^\circ\mathrm{C}))$$

Specific conductivities and temperature coefficients

Material	**Specific conductivity** $[\kappa] = \mathrm{S/m}$	**Temperature coefficient** $[\alpha] = 1/\mathrm{K}$
Silver	$6.1 \cdot 10^{7}$	0.0038
Copper	$5.8 \cdot 10^{7}$	0.0038
Gold	$4.5 \cdot 10^{7}$	0.0034
Aluminium	$3.7 \cdot 10^{7}$	0.004
Iron	$1.0 \cdot 10^{7}$	0.0065
Graphite	$3 \cdot 10^{6}$	-0.0002
Silicon (doped)	$1 - 10^{6}$	-0.075
Tap water	$5.0 \cdot 10^{-3}$	-
Air	$4.0 \cdot 10^{-15}$	-

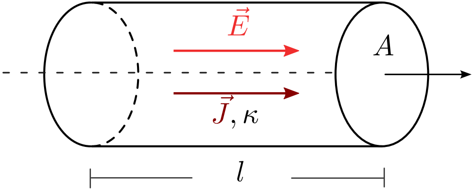
The conductance value

$$[G] = 1\,\mathsf{Siemens} = 1\,\mathsf{S} = 1\ \frac{1}{\Omega}$$

$$G = \frac{1}{R}$$

Geometric properties

- Length l
- Cross-sectional area A
- Specific conductivity κ



- General formula for resistance in relation to material properties

$$[R] = 1\,\text{Ohm} = 1\,\Omega$$

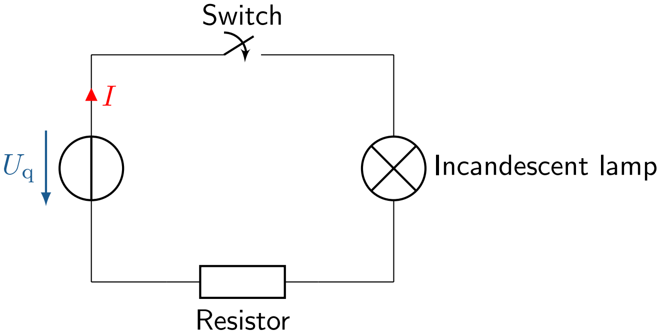
$$R = \rho_{\mathrm{R}} \cdot \frac{l}{A}$$

ρ_{R} : specific resistance of the material (Ωm)

l : Length of the material (m)

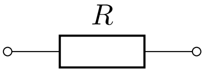
A : Cross-sectional area of the material (m^2)

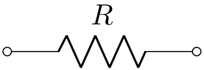
Introduction to the representation of circuits



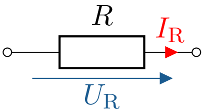
Resistance as a component

- European (left) and American (right) switching element for resistance

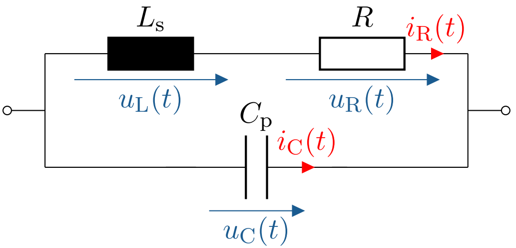




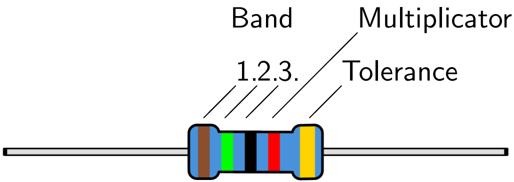
- The ideal resistor as a circuit symbol



- Equivalent circuit diagram of a real resistor with C_p and L_s



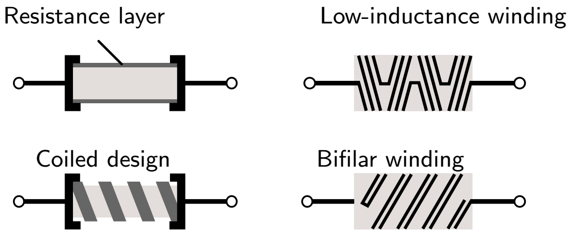
- Resistance color coding



$$1 - 5 - 0 - 2 = 150 \cdot 100\,\Omega = 15\,\mathrm{k}\Omega \pm 5\%$$

Colour	1. 2. 3. Band	Multiplicator	Tolerance
Black	0	$1\,\Omega$	-
Brown	1	$10\,\Omega$	$\pm 1\,\%$
Red	2	$100\,\Omega$	$\pm 2\,\%$
Orange	3	$1\,\mathrm{k}\Omega$	-
Yellow	4	$10\,\mathrm{k}\Omega$	-
Green	5	$100\,\mathrm{k}\Omega$	$\pm 0.5\,\%$
Blue	6	$1\,\mathrm{M}\Omega$	$\pm 0.25\,\%$
Violette	7	$10\,\mathrm{M}\Omega$	$\pm 0.1\,\%$
Grey	8	-	$\pm 0.05\,\%$
White	9	-	-
Gold	-	$0.1\,\Omega$	$\pm 5\,\%$
Silver	-	$0.01\,\Omega$	$\pm 10\,\%$

- Various designs of through-hole/THT resistors



(a) Layer resistors

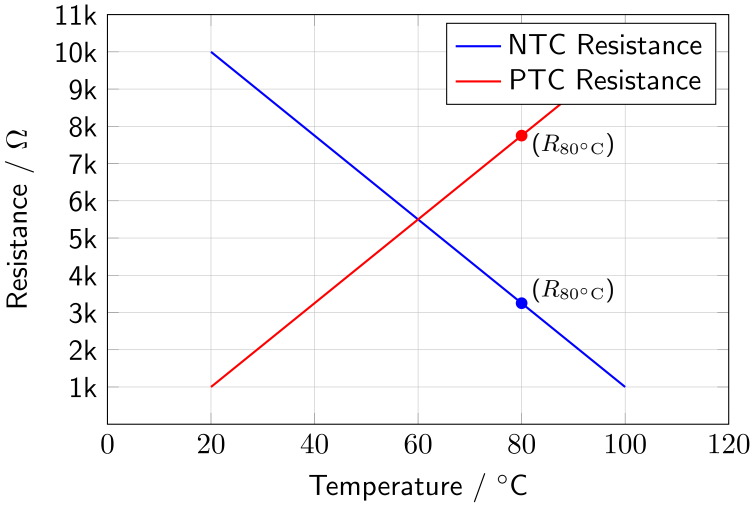
(b) Wirewires

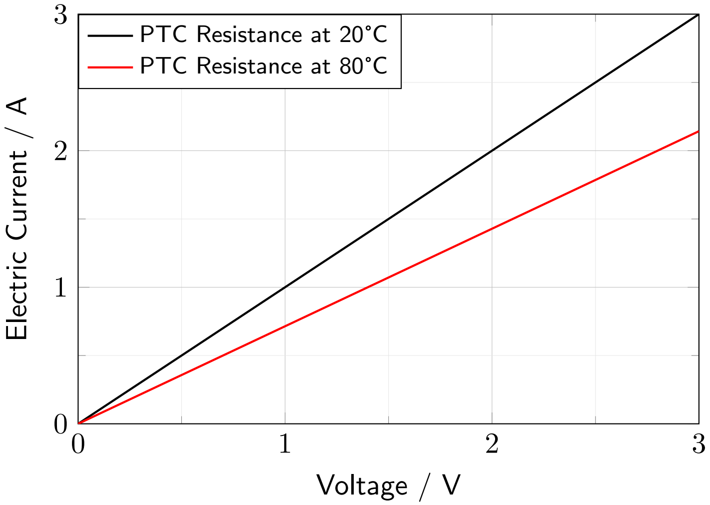
Mnemonic: Resistance as a component

Key point:

Der Widerstandswert eines Materials hängt von seinen geometrischen und spezifischen Eigenschaften ab.

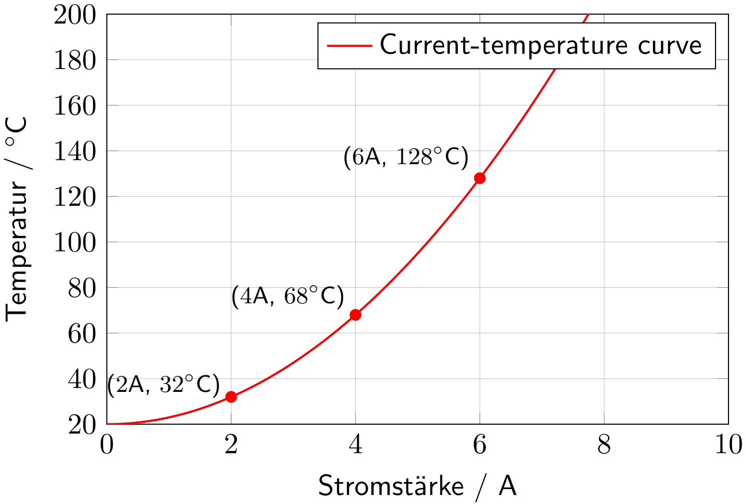
Temperature-dependent resistors





PTC Resistance at 20°C
PTC Resistance at 80°C
Electric Current / A
Voltage / V
0
1
2
3
0
1
2
3

Temperature-dependent resistors



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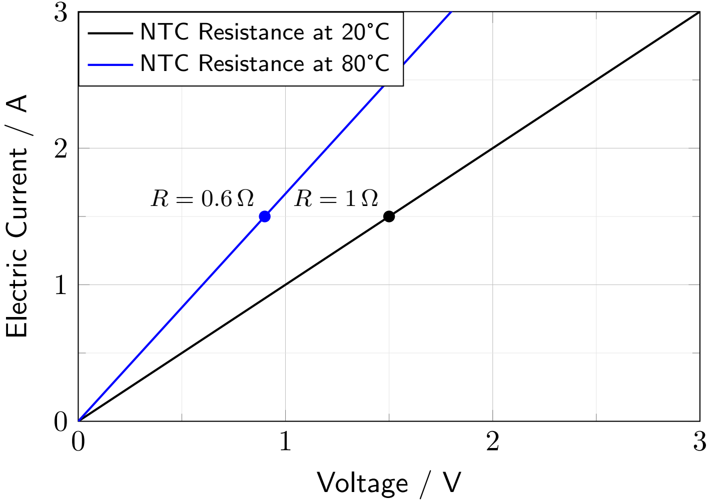
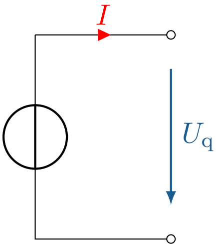


Learning objectives: Voltage and current source

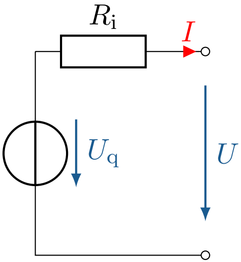
The Students can

- distinguish between voltage and current sources and know their circuit symbols.
- model real voltage and current sources.
- name examples of different sources.

- Ideal voltage source, always supplies a constant voltage U_q, regardless of the load or current



- Real voltage source. The voltage U_q is influenced by the internal resistance R_i and is therefore lower at the output terminals.



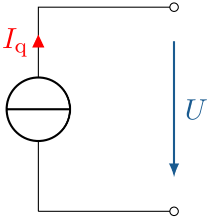
Mnemonic: voltage source

Key point:

- A voltage source must never be short-circuited!
- With voltage sources, the current always adjusts itself.

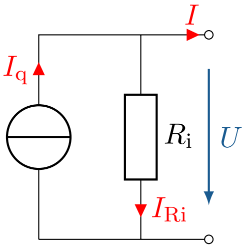
Current source

- Ideal current source



Current source

- Real current source with internal resistance. The current I_q is influenced by the internal resistance R_i and is lower at the output terminals.



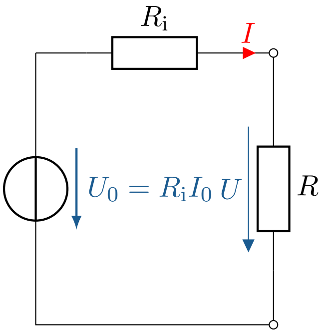
$$I = I_q - I_{R_i} = I_q - \frac{U}{R_i}$$

Mnemonic: Current source

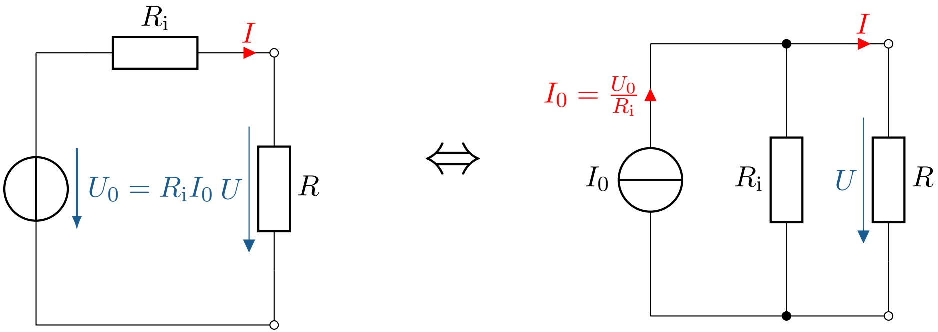
Key point:

- A power source must never be operated in idle mode!
- With power sources, the voltage always adjusts itself.

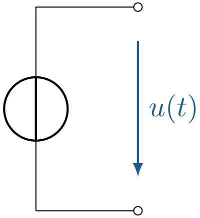
Conversion of a voltage source into a current source



Conversion of a voltage source into a current source

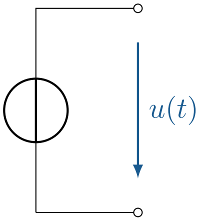


Various voltage and current sources



General
Voltage source

Various voltage and current sources



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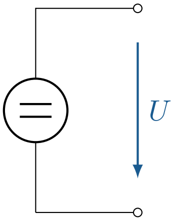
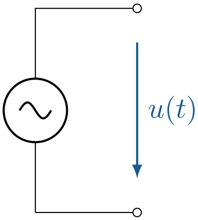
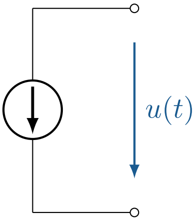
Voltage source



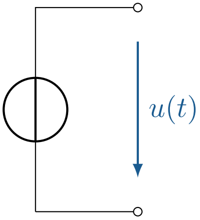
DC voltage
source



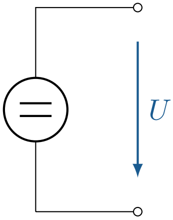
AC voltage
source



Temporally
variable voltage
source



General
Voltage source



DC voltage
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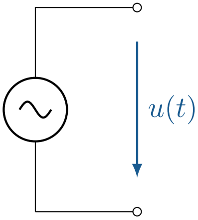
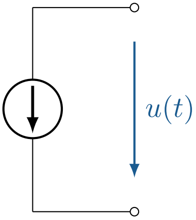
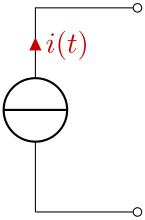


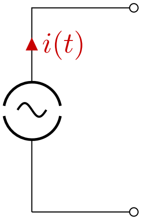
AC voltage
source



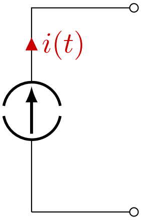
Temporally
variable voltage
source



General



AC source



Time-variable

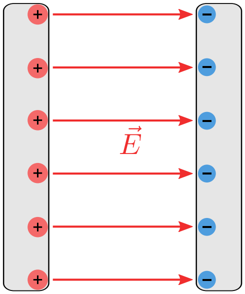
Learning objectives: Capacity and capacitor

Learning objectives: Capacity and capacitor

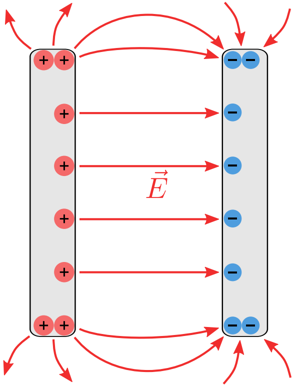
The Students

- can differentiate between capacity and capacitor.
- are familiar with the most important parameters relating to capacitors and capacitance.
- can calculate the capacity of a capacitor.

- The ideal plate capacitor



- The real plate capacitor



Electrical capacity

$$[C] = 1\,\mathsf{Farad} = 1\,\mathsf{F} = 1\,\frac{\mathsf{C}}{\mathsf{V}} = 1\,\frac{\mathsf{As}}{\mathsf{V}}$$

$$C = \frac{Q}{U}$$

Electrical capacity

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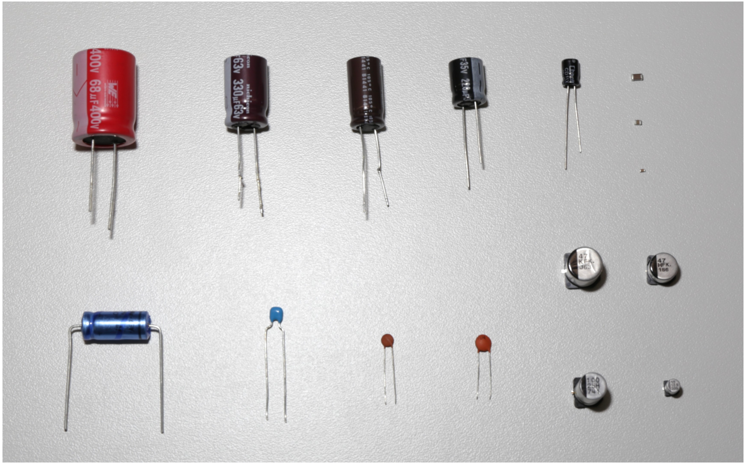
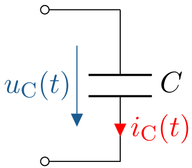
$$C = \frac{Q}{U}$$

$$Q = C \cdot U \quad \Rightarrow \quad C = \frac{Q}{U} = \frac{\varepsilon \cdot \iint \vec{E} \cdot \mathrm{d}\vec{A}}{\int \vec{E} \mathrm{d}\vec{s}}$$

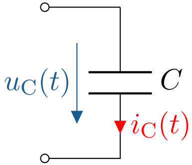
- Different types and designs of capacitors



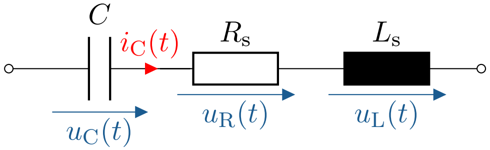
- Ideal capacitor



- Ideal capacitor



- Equivalent circuit diagram of a real capacitor



For a realistic simulation of the circuit, it is important to describe the capacitor as a component in its entirety. This is done with the aid of an equivalent circuit diagram, which also simulates the unwanted, i.e. parasitic, properties of the component. Figure 1 shows such an equivalent circuit diagram.

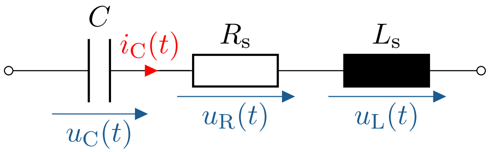


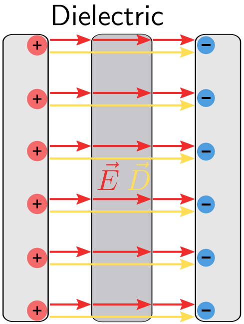
Figure: **Equivalent circuit diagram of a real capacitor.** The ideal capacitor C is supplemented by a series inductance L_{s} and a series resistance R_{s} to account for parasitic effects.

Key point: The capacitor as a component

Key point:

The capacitor is a desperate attempt to replicate a capacitance.

The dielectric



The dielectric

- Relative permittivity of various materials

Material	$\varepsilon_\mathbf{r}$
Vacuum	1
Air (at STP)	1.0006
Plastic (PE)	2.25-2.3
Glass	3-10
Ceramics	5-300
Silicon	11.7
Tantalum oxide	25-30
Water	81

The dielectric

$$[\varepsilon] = 1\,\frac{\text{Farad}}{\text{m}} = 1\,\frac{\text{F}}{\text{m}} = 1\,\frac{\text{As}}{\text{Vm}}$$

$$\varepsilon = \varepsilon_0 \cdot \varepsilon_\mathrm{r}$$

ε : Dielectric constant

ε_0 : Electric field constant ($8.85421878 \cdot 10^{-12}$)

ε_r : Relative permittivity

The capacitor as a component

$$[C] = 1\,\mathsf{F}$$

$$C = \frac{\varepsilon \cdot A}{d}$$

A : Cross-sectional area of the material(m^2)

d : Distance between electrodes (m)

The electric flux density $\vec{D}$

$$\vec{D} = \varepsilon \cdot \vec{E} + P$$

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- In static cases, polarization P can often be neglected.

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$$[D] = 1\,\frac{\text{Coulomb}}{\text{m}^2} = 1\,\frac{\text{C}}{\text{m}^2} = 1\,\frac{\text{As}}{\text{m}^2}$$

$$\vec{D} = \varepsilon \cdot \vec{E}$$

- $$Q = C \cdot U$$

Introduction to time-dependent processes

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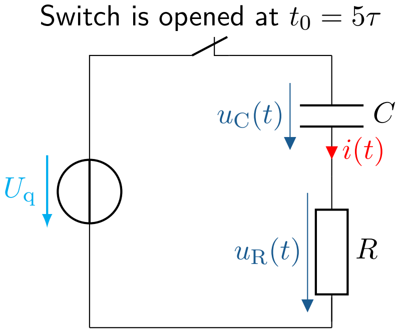
Switching behavior of a capacitor: Charging

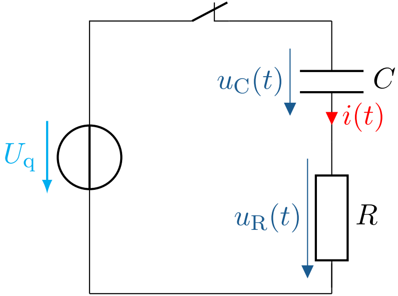
- $I_{\max} = \dfrac{U_q}{R}$

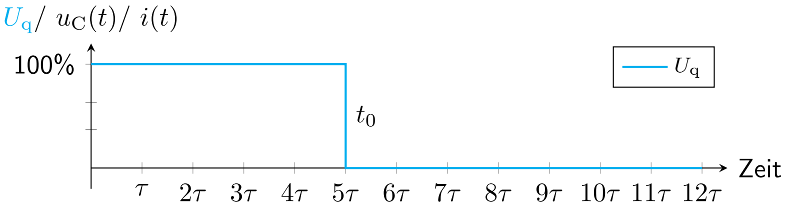


Switching behavior of a capacitor: Charging

- $I_{\max} = \dfrac{U_q}{R}$

Switch is opened at $t_0 = 5\tau$





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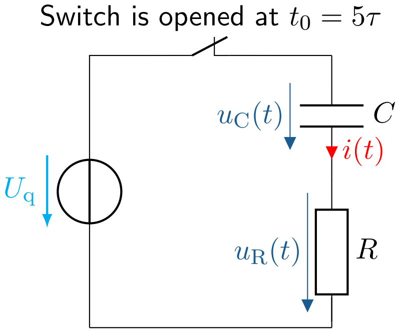

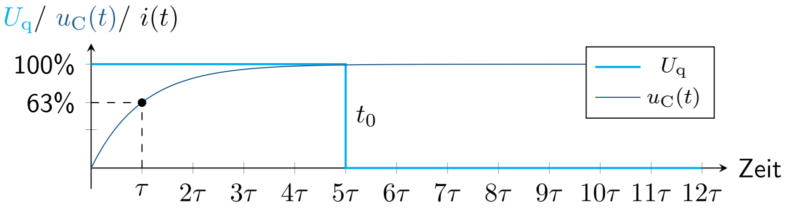

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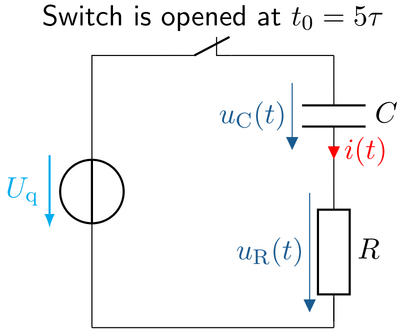
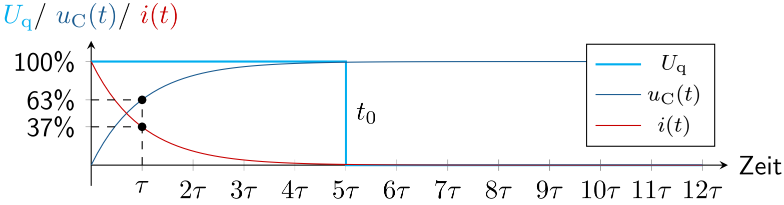
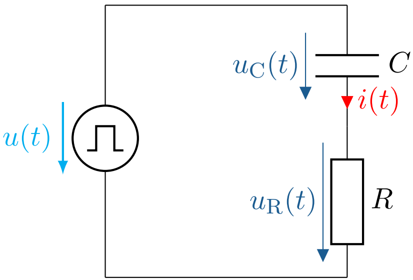
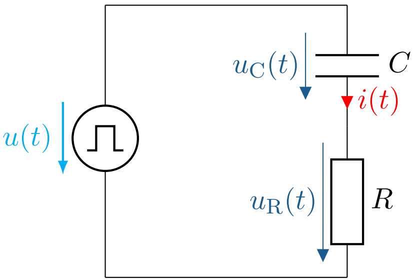
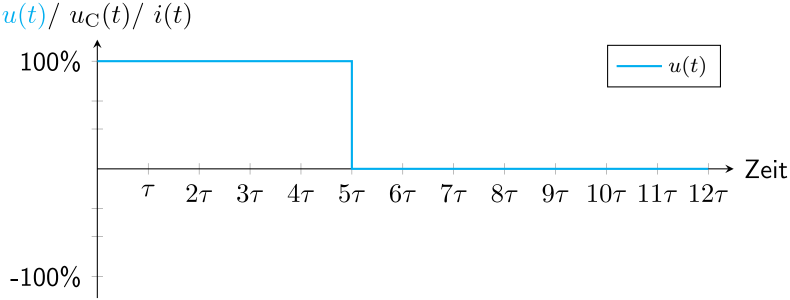




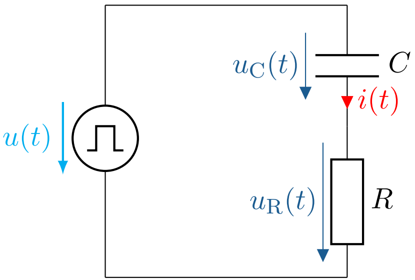
Switching behavior of a capacitor: Discharging

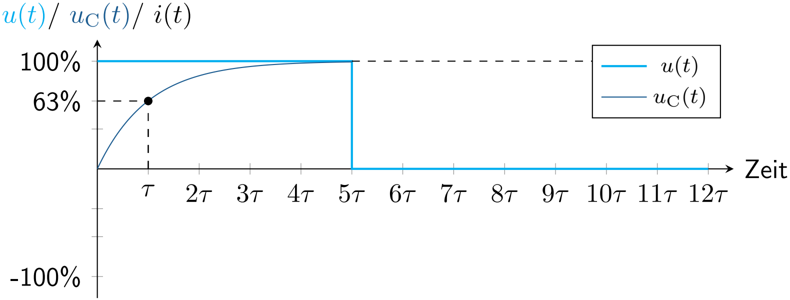



$u_{\mathrm{C}}(t)$
C
$i(t)$
$u(t)$
$u_{\mathrm{R}}(t)$
R

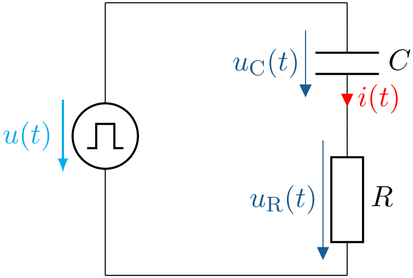

$u(t)$/ $u_{\mathrm{C}}(t)$/ $i(t)$
100%
-100%
Zeit
τ 2τ 3τ 4τ 5τ 6τ 7τ 8τ 9τ 10τ 11τ 12τ
$u(t)$

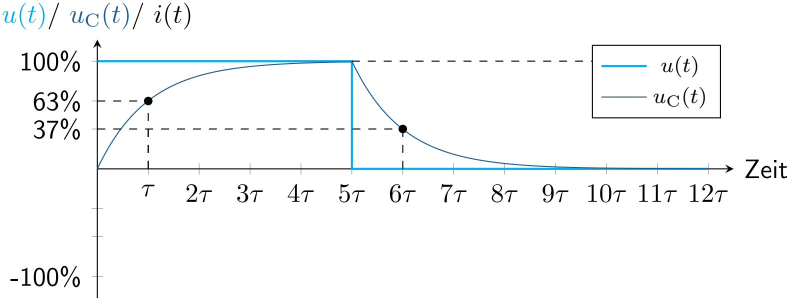
Switching behavior of a capacitor: Discharging



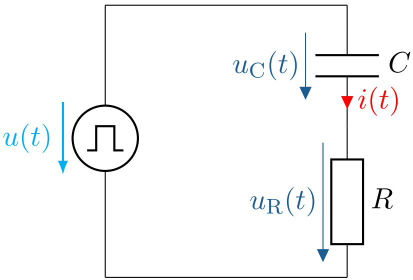


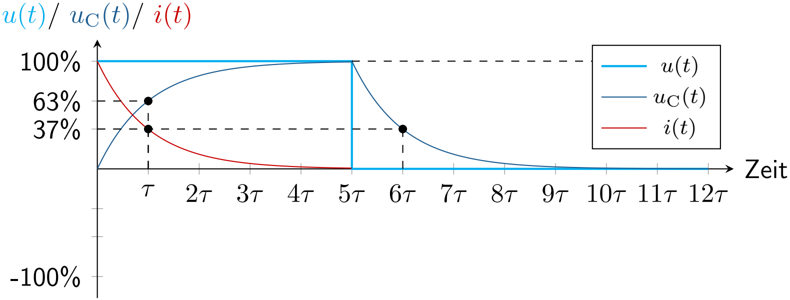
Switching behavior of a capacitor: Discharging



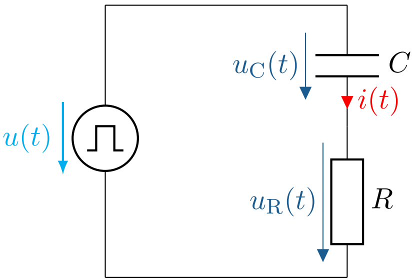


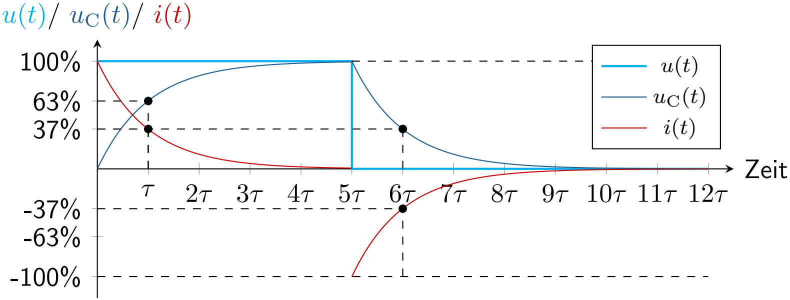
Switching behavior of a capacitor: Discharging





Switching behavior of a capacitor: Discharging





Sample calculation of capacity

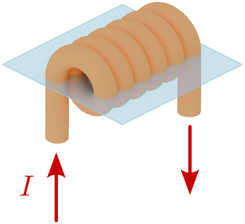
- Charge on plates $Q = \sigma \cdot A = \varepsilon \cdot E \cdot A$
- Calculate capacitance with $C = \frac{\varepsilon \cdot A}{d}$
- Stored energy $W = \frac{1}{2} \cdot C \cdot V^2$
- Calculate the capacitance of two capacitors with different ε_r over $\vec{D}$

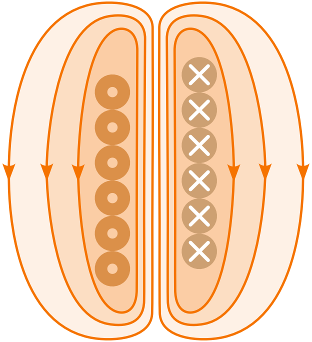
Learning objectives: Inductance and coil

The Students

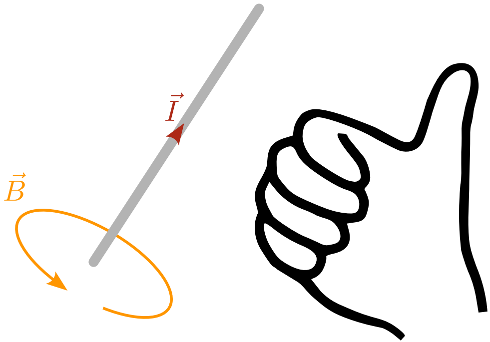
- can differentiate between inductance and coils.
- know the most important parameters relating to inductance and coils.
- can calculate the inductance of a coil.

Magnetic field of a cylindrical air coil





The right-hand rule



Induced voltage

$$U_\mathrm{i} \sim \frac{\mathrm{d}\phi}{\mathrm{d}t}$$

$$U_\mathrm{i} = -N \cdot \frac{\mathrm{d}\phi}{\mathrm{d}t}$$

$$U_\mathrm{i} \sim \frac{\mathrm{d}I}{\mathrm{d}t}$$

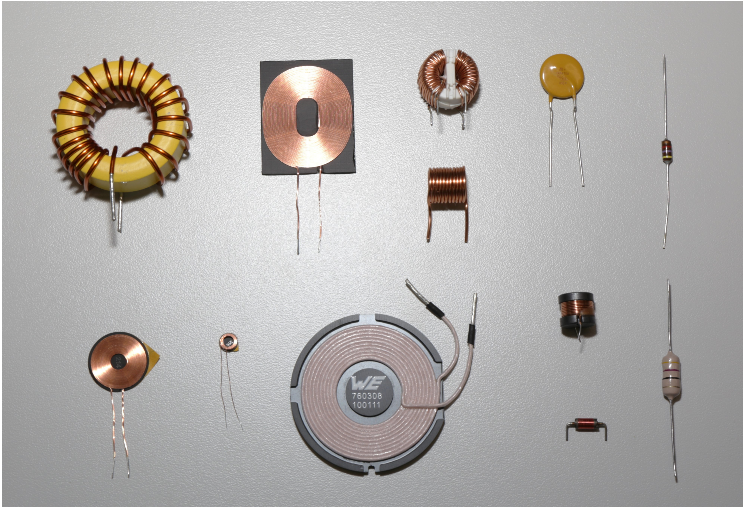
$$U_\mathrm{i} = -L \cdot \frac{\mathrm{d}I}{\mathrm{d}t}$$

- Voltage across the inductor

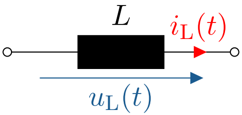
$$u_{\mathrm{L}}(t) = L \cdot \frac{di_{\mathrm{L}}(t)}{dt}$$

- Current through the inductance

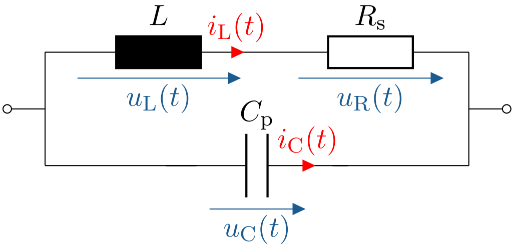
$$i_{\mathrm{L}}(t) = \frac{1}{L} \cdot \int u_{\mathrm{L}}(t)\, dt$$


WE
760308
100111

Inductance as a switching element



The coil as a switching element

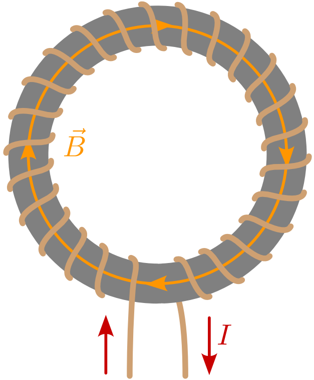


Key phrase: The coil as a component

Key point:

The coil is a desperate attempt to replicate inductance.

Permeability



Permeability

Material	μ_r
Type I superconductor	0
Vacuum	£1
Air (at STP)	£1.00000037
Copper	£0.999994
Gold	£0.999964
Aluminium	£1.000022
Iron	$300 - 10000$
Ferrite	$4 - 15000$

Permeability

$$\mu = \mu_\mathrm{r} \cdot \mu_0 \quad , \quad [\mathsf{H/m}]$$

μ : Permeability constant [H/m]

μ_0 : Magnetic field constant ($\approx 4\pi \times 10^{-7}$ [H/m])

μ_r : Relative permeability of the material (Einheitenfrei)

The magnetic flux density

$$B = \frac{\mu \cdot I \cdot N}{l}$$

$$[B] = \frac{\mathrm{H/m} \cdot \mathrm{A} \cdot 1}{\mathrm{m}} = \frac{\mathrm{H} \cdot \mathrm{A}}{\mathrm{m}^2} = \frac{(\mathrm{V} \cdot \mathrm{s}\ /\ \cancel{\mathrm{A}}) \cdot \cancel{\mathrm{A}}}{\mathrm{m}^2} = \frac{\mathrm{V} \cdot \mathrm{s}}{\mathrm{m}^2} = \mathrm{T}\ (\mathrm{Tesla})$$

μ : Permeability constant]

I : Electrical current

N : Number of turns

l : Length of the magnetic circuit

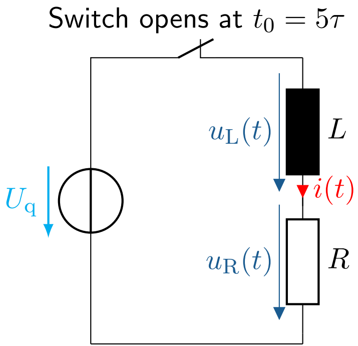
$$\vec{B} = \mu \cdot \vec{H}$$

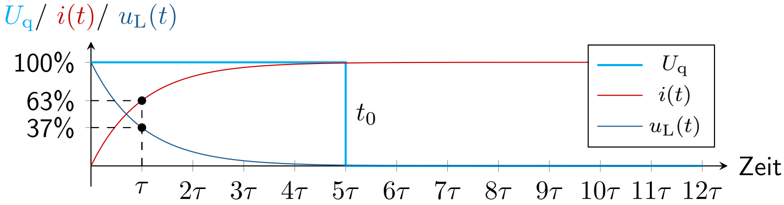
The inductance

$$[L] = 1\,\mathsf{Henry} = 1\,\mathsf{H} = 1\,\frac{\mathsf{Vs}}{\mathsf{A}}$$

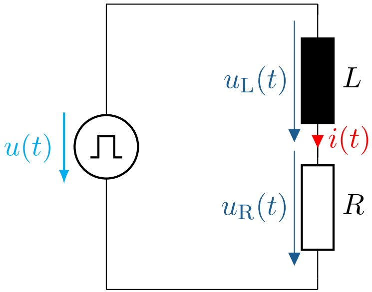
$$L = \frac{\mu \cdot N^2 \cdot A}{l}$$

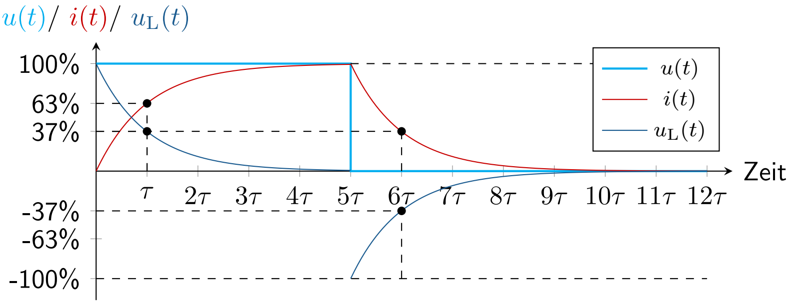
Switching behavior of a coil: Charging





Switching behavior of a coil: Discharging





Example calculation for inductance

- Calculation of inductance L
- Calculation of magnetic flux density $\vec{B}$
- Energy stored in the magnetic field E_m ?